

# Xiao FU (付潇)

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## EDUCATION

### The Chinese University of Hong Kong (CUHK)

Ph.D. in Information Engineering, Multimedia Laboratory (MMLab)

• **Advisor:** Prof. *Dahua Lin*, **HKPFS Awardee**

Hong Kong SAR

Sept. 2023 – June 2027 (Expected)

### Zhejiang University (ZJU)

B.Eng. (Honor) in Information Engineering

• **Adivisor:** Prof. *Yiyi Liao*, **GPA:** 3.96/4.00, **Ranking:** 1/144

Hangzhou, China

Sept. 2018 – Jul. 2022

## WORKING EXPERIENCE

### NVIDIA Research

Research Scientist Intern. Adivisor : Dr. *Ming-Yu Liu* (VP of Research)

Topic: Multi-Modality, Video Foundation Model

Santa Clara, United States

May 2025 – Present

### Kling, Kuaishou VGI Lab

Research Intern. Adivisor : Dr. *Xintao Wang*

Topic: Controllable Video Generation

Shenzhen, China

May 2024 – May 2025

### Shanghai AI Innovation Center

Associate Researcher. Adivisor : Prof. *Dahua Lin*

Topic: OmniObject3D: A Large-Vocabulary and Realistic 3D Object Dataset

Shanghai, China

Jul. 2022 – May. 2023

### Max Planck Institute for Intelligent Systems

Researcher Assistant. Adivisor : Prof. *Andreas Geiger*, Prof. *Yiyi Liao*

Topic: Panoptic Neural Radiance Field for 3D-2D Label Transfer

Tübingen, Germany (remote)

Sept. 2021 – Mar. 2022

## SELECTED PUBLICATION

### (1) Multi-Modal Image/Video Generation

#### • Learning Video Generation for Robotic Manipulation with Collaborative Trajectory Control

Xiao Fu, Xintao Wang, Xian Liu, Jianhong Bai, Runsen Xu, Pengfei Wan, Di Zhang, Dahua Lin

ARXIV 2025 (Under Review of **NeurIPS**, 2025) | [website](#) | [code](#)

#### • 3DTrajMaster: Mastering 3D Trajectory for Multi-Entity Motion in Video Generation

Xiao Fu, Xian Liu, Xintao Wang, Sida Peng, Menghan Xia, Xiaoyu Shi, Ziyang Yuan, Pengfei Wan, Di Zhang, Dahua Lin

International Conference on Learning Representations (ICLR), 2025 | [website](#) | [dataset](#) | [code](#) (300+ stars)

#### • GeoWizard: Unleashing the Diffusion Priors for 3D Geometry Estimation from a Single Image

Xiao Fu, Wei Yin, Mu Hu, Kaixuan Wang, Yuexin Ma, Ping Tan, Shaojie Shen, Dahua Lin, Xiaoxiao Long

European Conference on Computer Vision (ECCV), 2024 | [website](#) | [code](#) (800+ stars) (Citation: 100+)

#### • ReCamMaster: Camera-Controlled Generative Rendering from A Single Video

Jianhong Bai, Menghan Xia, Xiao Fu, Xintao Wang, Lianrui Mu, Jinwen Cao, Zuozhu Liu, Haoji Hu, Xiang Bai, Pengfei Wan, Di Zhang

International Conference on Computer Vision (ICCV), 2025 (Oral Presentation) | [website](#) | [code](#) (1.3K+ stars)

#### • SynCamMaster: Synchronizing Multi-Camera Video Generation from Diverse Viewpoints

Jianhong Bai, Menghan Xia, Xintao Wang, Ziyang Yuan, Xiao Fu, Zuozhu Liu, Haoji Hu, Pengfei Wan, Di Zhang

KwaiVGI's Multi-Camera VideoGen Project, 2025 | [website](#) | [dataset](#) | [code](#) (600+ stars)

### (2) 3D Generation & Perception

#### • PanopticNeRF-360: Panoramic 3D-to-2D Label Transfer in Urban Scenes

Xiao Fu, Shangzhan Zhang, Tianrun Chen, Yichong Lu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao

IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI), 2025 | [website](#) | [code](#) (200+ stars)

#### • Panoptic NeRF: 3D-to-2D Label Transfer for Panoptic Urban Scene Segmentation

Xiao Fu, Shangzhan Zhang, Tianrun Chen, Yichong Lu, Lanyun Zhu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao

International Conference on 3D Vision (3DV), 2022 | [website](#) | [code](#) (200+ stars) (Citation: 100+)

- **OmniObject3D: Large-Vocabulary 3D Object Dataset for Realistic Perception, Reconstruction and Generation**

Tong Wu, Jiarui Zhang, **Xiao Fu**, Yuxin Wang, Jiawei Ren, Liang Pan, Wayne Wu, Lei Yang, Jiaqi Wang, Chen Qian, Dahua Lin, Ziwei Liu

Computer Vision and Pattern Recognition (CVPR), 2023 (**Best Paper Award Candidate, Top 12/9,155**) | [code](#)

## INVITED TALK & WORKSHOPS

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- **AnySyn3D Webinar:** 3D Interactive Video Generation 2024
- **AI for 3D Content Creation Workshop at ICCV 2023:** OmniObject3D Challenge 2023
- **Shenlan Open Courses:** Panoramic Label Rendering for Autonomous Driving Simulator 2023

## HONORS & AWARDS

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- CVPR Best Paper Award Candidate 2023
- Hong Kong PhD Fellowship Scheme (HKPFS) (**Top 300 Ph.D. students in Hong Kong**) 2023
- CUHK Vice-Chancellor HKPFS Scholarship 2023
- Outstanding Graduation Thesis Award of Zhejiang University 2022
- National Scholarship, Ministry of Education of P.R. China (**Top 0.2% across China**) 2020,2021
- ZJU ISEE Excellent Student Award (**Dean's Honor Undergraduate**) 2021
- ZJU ISEE Yuelun Alumni Scholarship (**Dean's Award for Academic Contest**) 2020
- Finalist Winner, Mathematical Contest in Modeling (MCM/ICM) (**Top 2% out of 20,954 teams**) 2020
- Gold Prize, 7<sup>th</sup> "Internet Plus" College Student Innovation and Entrepreneurship Contest (1<sup>st</sup> **author**) 2021